DPEN022: Enabling Mathematics B
Class Questions Week 4

Week 4A

Key Idea – Derivatives of Logarithmic Functions

1. (a) $y' = \dfrac{4}{4x}$
$= \dfrac{1}{x}$

(b) $y' = 3 \times \dfrac{6}{6x+5} + 2$
$= \dfrac{18}{6x+5} + 2$

(c) $f(x) = \log_e(2x^4+1)$
$f'(x) = \dfrac{8x^3}{(2x^4+1)}$

(d) $y' = \dfrac{4}{4x+3} - 2$

(e) $y' = \dfrac{2}{x} + 6x$

(f) $f'(x) = 2 \times \dfrac{2x-4}{x^2-4x+4}$
$= \dfrac{4x-8}{x^2-4x+4}$

(g) $f'(x) = \dfrac{\frac{1}{2\sqrt{x}}}{1+\sqrt{x}}$
$= \dfrac{1}{2\sqrt{x}\left(1+\sqrt{x}\right)}$

(h) $y' = \dfrac{4x^3}{x^4} + 3$
$= \dfrac{4}{x} + 3$

(i) $y = \dfrac{\frac{1}{2}}{\frac{x}{2}}$
$= \dfrac{1}{x}$

2. (a) $y = \log_e(x+2)(x-5)$
$= \log_e(x+2) + \log_e(x-5)$
$y' = \dfrac{1}{x+2} + \dfrac{1}{x-5}$

(b) $f(x) = \log_e\sqrt{(x^2+4)} = \log_e(x^2+4)^{1/2}$
$= \dfrac{1}{2}\log_e(x^2+4)$
$f'(x) = \dfrac{1}{2} \times \dfrac{2x}{x^2+4}$
$= \dfrac{x}{x^2+4}$

(c) $f(x) = \log_e\dfrac{(x+2)}{(x-1)}$
$= \log_e(x+2) - \log_e(x-1)$
$f'(x) = \dfrac{1}{x+2} - \dfrac{1}{x-1}$

(d) $g(x) = \log_e\dfrac{2}{(4x-3)}$
$= \log_e 2 - \log_e(4x-3)$
$g'(x) = 0 - \dfrac{4}{4x-3}$
$= -\dfrac{4}{4x-3}$ or $\dfrac{4}{3-4x}$

(e) $f(x) = \log_e 2x\left(x^2-1\right)$
$f'(x) = \log_e 2x + \log_e\left(x^2-1\right)$
$= \dfrac{2}{2x} + \dfrac{2x}{x^2-1}$
$= \dfrac{1}{x} + \dfrac{2x}{x^2-1}$

(f) $y = \log_e\left(3x^2+1\right)^2$
$= 2\log_e\left(3x^2+1\right)$
$y' = 2 \times \dfrac{6x}{3x^2+1}$
$= \dfrac{12x}{3x^2+1}$

3. Find the derivative of the following *[using product, quotient and chain rule]*:

(a) $y = 2x^3 \ln x$ $\quad u = 2x^3$, $u' = 6x^2$, $v = \ln x$, $v' = \dfrac{1}{x}$

$$y' = 6x^2 \ln x + 2x^3 \frac{1}{x}$$
$$= 6x^2 \ln x + 2x^2$$

(b) $y = (x+3)\ln(x+3)$ $\quad u = (x+3)$, $u' = 1$, $v = \ln(x+3)$, $v' = \dfrac{1}{x+3}$

$$y' = (x+3) \times \frac{1}{x+3} + \ln(x+3) \times 1$$
$$= 1 + \ln(x+3)$$

(c) $y = \dfrac{\log_e(x^2+1)}{x}$ $\quad u = \log_e(x^2+1)$, $u' = \dfrac{2x}{x^2+1}$, $v = x$, $v' = 1$

$$y' = \frac{x \times \frac{2x}{x^2+1} - \log_e(x^2+1) \times 1}{x^2}$$
$$= \frac{\frac{2x^2}{x^2+1} - \log_e(x^2+1)}{x^2}$$

(d) $f(x) = \left[\log_e(x^3+2)\right]^4$

$$= 4\left[\log_e(x^3+2)\right]^3 \times \left(\frac{3x^2}{x^3+2}\right)$$
$$f'(x) = \frac{12x^2}{x^3+2}\left[\log_e(x^3+2)\right]^3$$

(e) $g(x) = 1 + 2x\log_e x$ $\quad u = 2x$, $u' = 2$, $v = \log_e x$, $v' = \dfrac{1}{x}$

$$g'(x) = 0 + 2\log_e(x) + 2x \times \frac{1}{x}$$
$$= 2\log_e(x) + 2$$

(f) $y = 2x - \left[\log_e(x+2)\right]^3$

$$y' = 2 - -3\left[\log_e(x+2)\right]^2 \times \frac{1}{x+2}$$
$$= 2 - \frac{3}{x+2}\left[\log_e(x+2)\right]^2$$

4. (a) $g(x) = x^2 \log_e x$ $\quad u = x^2$, $u' = 2x$, $v = \log_e x$, $v' = \dfrac{1}{x}$

$$g'(x) = 2x\log_e x + x^2 \times \frac{1}{x}$$
$$= 2x\log_e x + x$$
$$g'(1) = 2 \times 1 \times \log_e 1 + 1$$
$$= 1$$

(b) $y = x\log_e(x) - x$ $\quad u = x$, $u' = 1$, $v = \log_e x$, $v' = \dfrac{1}{x}$

$$y' = \left[\log_e(x) + x \times \frac{1}{x}\right] - 1$$
$$= \log_e(x) + 1 - 1$$
$$= \log_e x$$
$$y'' = \frac{1}{x}$$

(c) $y = \dfrac{\ln x}{x}$ $\quad u = \ln x$, $u' = \dfrac{1}{x}$, $v = x$, $v' = 1$

$$y' = \frac{x \times \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}$$

$$y' = \frac{1 - \ln x}{x^2} = 0 \quad \text{when } 1 - \ln x = 0$$

$\ln x = 1$

$x = e$

Therefore, the point on the curve is when $x = e$ and $y = \dfrac{\ln e}{e} = \dfrac{1}{e} = e^{-1}$

Point is (e, e^{-1})

(d) $f'(x) = \dfrac{1}{x-1} \rightarrow f'(2) = \dfrac{1}{2-1} = 1$

Find the y value at $x = 2$ → $f(2) = \log_e 1$

$$= 0$$

$$\therefore \textit{point is } (2,\ 0)$$

$$y - y_1 = m(x - x_1)$$
$$y - 0 = 1(x - 2)$$
$$y = x - 2$$
$$x - y - 2 = 0$$

Key Idea – Derivatives of Trigonometric Functions

1. (a) $y' = 3\cos 3x$ (b) $f'(x) = \cos x - 4\sin x$ (c) $y' = 2x\sec^2 x^2$

(d)

$$y = 2\sin(2x)$$
$$\frac{dy}{dx} = 2\times 2\cos(2x)$$
$$= 4\cos(2x)$$

(e)

$$y = \tan(4x-1)$$
$$y' = 4\sec^2(4x-1)$$

(f)

$$y = 4\cos\left(x + \frac{\pi}{3}\right)$$
$$\frac{dy}{dx} = 4\times\left(-\sin\left(x+\frac{\pi}{3}\right)\right)$$
$$= -4\sin\left(x + \frac{\pi}{3}\right)$$

(g) $f'(x) = 6\cos 2x + 2\sin 2x$ (h) $y' = 4\sec^2\left(\frac{\pi}{4} + 4x\right) - 4x$

(i) $y' = -\frac{1}{2}\cos\frac{x}{2} - 3x^2$

2. (a)

$$y = x^2\sin(2x)$$

Product Rule $(y = u\times v)$

$u = x^2 \quad v = \sin(2x)$

$u' = 2x \quad v' = 2\cos(2x)$

$$y' = vu' + uv'$$
$$= \sin(2x)\times 2x + x^2\times 2\cos(2x)$$
$$= 4x\sin(2x) + 2x^2\cos(2x)$$
$$= 2x\left(2\sin(2x) + x\cos(2x)\right)$$

(b)

$$y = \sqrt{(1+\tan(3x))}$$
$$= (1+\tan(3x))^{\frac{1}{2}}$$

Chain Rule

$$\frac{dy}{dx} = \frac{1}{2}(1+\tan(3x))^{-\frac{1}{2}}\times 3\sec^2(3x)$$
$$= \frac{3\sec^2(3x)}{2\sqrt{(1+\tan(3x))}}$$

(c)

$$y = \sin^2(2x+1)$$
$$= (\sin(2x+1))^2$$

Chain Rule

$$y' = 2(\sin(2x+1))^1\times 2\cos(2x+1)$$
$$= 4\sin(2x+1)\cos(2x+1)$$

(d)

$$y = \frac{x^2}{\cos x}$$

Quotient Rule

$u = x^2 \quad v = \cos x$

$u' = 2x \quad v' = -\sin x$

$$y' = \frac{vu' - uv'}{v^2} = \frac{\cos x 2x + x^2\sin x}{x^4}$$
$$= \frac{x(2\cos x + x\sin x)}{x^4}$$
$$= \frac{2\cos x + x\sin x}{x^3}$$

(e) $y' = \cos\frac{x}{2} - x \times \frac{1}{2}\sin\frac{x}{2}$

$= \cos\frac{x}{2} - \frac{x}{2}\sin\frac{x}{2}$

(f) $y' = \sin x(1+\cos x)$

$= (1+\cos x)\cos x + \sin x(-\sin x)$

$= \cos x + \cos^2 x - \sin^2 x$

$= \cos x(1+\cos x) - \sin^2 x$

(g)

$y = \frac{x^3}{\cos x}$

Quotient Rule $\left(y = \frac{u}{v}\right)$

$u = x^3 \qquad v = \cos x$

$u' = 3x^2 \qquad v' = -\sin x$

$y' = \frac{vu' - uv'}{v^2} = \frac{\cos x \times 3x^2 - x^3 \times (-\sin x)}{(\cos x)^2}$

$= \frac{3x^2\cos x + x^3 \sin x}{\cos^2 x}$

$= \frac{x^2(3\cos x + x\sin x)}{\cos^2 x}$

(h)

$y = \frac{x}{\tan(2x)}$

Quotient Rule $\left(y = \frac{u}{v}\right)$

$u = x \qquad v = \tan(2x)$

$u' = 1 \qquad v' = 2\sec^2(2x)$

$y' = \frac{vu' - uv'}{v^2}$

$= \frac{\tan(2x) \times 1 - x \times 2\sec^2(2x)}{(\tan(2x))^2}$

$= \frac{\tan(2x) - 2x\sec^2(2x)}{\tan^2(2x)}$

3. (a)

$f(x) = 2\cos(3x)$

$f'(x) = 2 \times -3\sin(3x)$

$= -6\sin(3x)$

$f'\left(\frac{\pi}{12}\right) = -6\sin\left(3 \times \frac{\pi}{12}\right)$

$= -6\sin\left(\frac{\pi}{4}\right)$

$= -6 \times \frac{1}{\sqrt{2}} = -\frac{6}{\sqrt{2}} = -3\sqrt{2}$

(b) $y = 2\sin x - 4\cos(4x-1)$

$\frac{dy}{dx} = 2\cos x - 4 \times (-4\sin(4x-1))$

$= 2\cos x + 16\sin(4x-1)$

$\frac{d^2y}{dx^2} = -2\sin x + 16 \times 4\cos(4x-1)$

$= -2\sin x + 64\cos(4x-1)$

(c)

$y = \tan x$

Find the gradient $\rightarrow\ y' = \sec^2 x$

Gradient of the tangent at $x = \dfrac{\pi}{3}$

$$m = y'\left(\frac{\pi}{3}\right) = \left(\sec\left(\frac{\pi}{3}\right)\right)^2 = 2^2 = 4$$

Find the y-coordinate when $x = \dfrac{\pi}{3} \rightarrow y = \tan\dfrac{\pi}{3} = \sqrt{3}$

Equation of the tangent using $m = 4$ and $(x_1, y_1) = \left(\dfrac{\pi}{3}, \sqrt{3}\right)$

$$y - y_1 = m(x - x_1) \rightarrow y - \sqrt{3} = 4\left(x - \frac{\pi}{3}\right)$$

$$y = 4x - \frac{4\pi}{3} + \sqrt{3}$$

Key Idea – Mixed Derivative of any Functions using Rules 1-5

1. (a) $y' = e^x(\sin x + \cos x)$ (b) $y' = e^{2x}\left(2\cos\frac{x}{2} - \frac{1}{2}\sin\frac{x}{2}\right)$ (c) $y' = e^{-x}(3\cos 3x - \sin 3x)$

(d) $y' = e^x(\cos 4x - 4\sin 4x)$ (e) $y' = -2e^{-x}(\sin x)$ (f) $y' = 2e^{\sin 2x}\cos 2x$

(g)

$$y' = e^x \ln x + e^x \frac{1}{x}$$

$$y' = e^x\left(\ln x + \frac{1}{x}\right)$$

(i)

$$\begin{aligned} y' &= e^x \log_e\left(e^x + 1\right) + e^x \frac{e^x}{e^x + 1} \\ &= e^x \log_e\left(e^x + 1\right) + \frac{e^{2x}}{e^x + 1} \\ &= e^x\left(\log_e\left(e^x + 1\right) + \frac{e^x}{e^x + 1}\right) \end{aligned}$$

(h)

$$y' = \frac{e^x \frac{1}{x} - e^x \log_e x}{\left(e^x\right)^2} \to \frac{\frac{e^x}{x} - e^x \log_e x}{\left(e^x\right)^2} \to \frac{e^x\left(\frac{1}{x} - \log_e x\right)}{\left(e^x\right)^2} \to \frac{\frac{1}{x} - \log_e x}{e^x}$$

2. (a)

$$f(x) = \left(2x - 3e^x + 4\tan x\right)^{11}$$

Use Chain Rule

$$\begin{aligned} f'(x) &= 11\left(2x - 3e^x + 4\tan x\right)^{10} \times \left(2 - 3e^x + 4\sec^2 x\right) \\ &= 11\left(2 - 3e^x + 4\sec^2 x\right)\left(2x - 3e^x + 4\tan x\right)^{10} \end{aligned}$$

(b)

$$f(x) = e^{x^2 + \tan(5x)}$$

$$f'(x) = \left(2x + 5\sec^2(5x)\right)e^{x^2 + \tan(5x)}$$

(c)

$$y = \tan\left(\sin x + \cos x\right)$$

$$y' = \left(\cos x - \sin x\right)\sec^2\left(\sin x + \cos x\right)$$

(d)

$$\begin{aligned} y &= \cos\left(e^{5x} + x\right) \\ \frac{dy}{dx} &= \left(5e^{5x} + 1\right) \times \left(-\sin\left(e^{5x} + x\right)\right) \\ &= -\left(5e^{5x} + 1\right)\sin\left(e^{5x} + x\right) \end{aligned}$$

(e)

$$\begin{aligned} y &= \cos x \tan x \\ &= \cos x \times \frac{\sin x}{\cos x} \\ &= \sin x \\ y' &= \cos x \end{aligned}$$

(f)

$$y = \ln\left(5x + \sin\left(x^2\right)\right)$$

$$y' = \frac{5 + 2x\cos\left(x^2\right)}{5x + \sin\left(x^2\right)}$$

(g)

$$\begin{aligned} y(x) &= \ln\left(\left(\cos x\right)^{-4}\right) \\ &= \frac{-4\left(\cos x\right)^{-3} \times \left(-\sin x\right)}{\left(\cos x\right)^{-4}} \\ &= 4\sin x\left(\cos x\right)^{-3}\left(\cos x\right)^{4} \\ &= 4\sin x \cos x \end{aligned}$$

(h)

$$f(x) = e^{\sin x \ln x}$$

$$f'(x) = \left(\cos x \ln x + \frac{\sin x}{x}\right)e^{\sin x \ln x}$$

Week 4D

Key Idea –The Primitive Function

1. Answer only $y = 3 + 6x - 2x^2$
2. Answer only $y = 2x^3 + 4x^2 - 9$

Key Idea – Integration of Power of *x* where *n* is an Integer

1. (a) $\frac{dy}{dx} = x^3$

$$y = \frac{x^{3+1}}{3+1} + c = \frac{x^4}{4} + c$$

(b) $\frac{dy}{dx} = 6$

$$y = 6x + c$$

(c) $\frac{dy}{dx} = 2x^4$

$$y = \frac{2x^{4+1}}{4+1} + c = \frac{2x^5}{5} + c$$

(d) $\frac{dy}{dx} = -6x^2$

$$y = \frac{-5x^{2+1}}{2+1} + c = \frac{-5x^3}{3} + c$$

(e) $\frac{dy}{dx} = \frac{4}{5}x^3$

$$y = \frac{4}{5} \times \frac{x^{3+1}}{3+1} + c = \frac{4}{5} \times \frac{x^4}{4} + c = \frac{x^4}{5} + c$$

(f) $\frac{dy}{dx} = \frac{x^2}{5}$

$$y = \frac{1}{5} \times \frac{x^{2+1}}{2+1} + c = \frac{1}{5} \times \frac{x^3}{3} + c = \frac{x^3}{15} + c$$

(g) $\frac{dy}{dx} = x^{-3}$

$$y = \frac{x^{-3+1}}{-3+1} + c = \frac{x^{-2}}{-2} + c = -\frac{1}{2x^2} + c$$

(h) $\frac{dy}{dx} = -2x^{-4}$

$$y = -\frac{2x^{-4+1}}{-4+1} + c = -\frac{2x^{-3}}{-3} + c = \frac{2}{3x^3} + c$$

(i) $\frac{dy}{dx} = \frac{2}{3}x^{-6}$

$$y = \frac{2}{3} \times \frac{x^{-6+1}}{-6+1} + c = \frac{2}{3} \times \frac{x^{-5}}{-5} + c = -\frac{2}{15x^5} + c$$

(j) $\frac{dy}{dx} = \frac{1}{x^4} = x^{-4}$

$$y = \frac{x^{-4+1}}{-4+1} + c = \frac{x^{-3}}{-3} + c = -\frac{1}{3x^3} + c$$

(k) $\frac{dy}{dx} = \frac{-3}{x^5} = -3x^{-5}$

$$y = \frac{-3x^{-5+1}}{-5+1} + c = \frac{3x^{-4}}{5} + c = \frac{3}{5x^4} + c$$

(l) $\frac{dy}{dx} = \frac{2}{3x^3} = \frac{2}{3}x^{-3}$

$$y = \frac{2}{3} \times \frac{x^{-3+1}}{-3+1} + c = \frac{2}{3} \times \frac{x^{-2}}{-2} + c = \frac{-1}{3x^2} + c$$

Key Idea – Integration of Power of x where n is a Fraction

1. Most of the solutions are answers only

(a) $\frac{4}{5}x^{\frac{5}{4}}+c$ index form

$\frac{4\sqrt[4]{x^5}}{5}+c$ surd form

(b) $\frac{10}{7}x^{\frac{7}{5}}+c$ index form

$\frac{10\sqrt[5]{x^7}}{7}+c$ surd form

(c) $-\frac{9}{7}x^{\frac{7}{3}}+c$ index form

$-\frac{9\sqrt[3]{x^7}}{7}+c$ surd form

(d) $\frac{4}{3}x^{\frac{3}{4}}+c$ index form

$\frac{4\sqrt[4]{x^3}}{3}+c$ surd form

(e) $32x^{\frac{1}{4}}+c$ index form

$32\sqrt[4]{x}+c$ surd form

(f) $2x^{\frac{1}{3}}+c$ index form

$2\sqrt[3]{x}+c$ surd form

(g) $\int\sqrt[3]{x^2}\,dx=\int x^{\frac{2}{3}}dx=\dfrac{x^{\frac{2}{3}+1}}{\frac{2}{3}+1}+C$

$=\dfrac{x^{\frac{5}{3}}}{\frac{5}{3}}+C=\dfrac{3\sqrt[3]{x^5}}{5}+C$

(h) $\int\sqrt{3x}\,dx=\int\sqrt{3}\times\sqrt{x}\,dx=\sqrt{3}\int x^{\frac{1}{2}}\,dx$

$=\sqrt{3}\times\dfrac{x^{\frac{3}{2}}}{\frac{3}{2}}+C=\sqrt{3}\times\dfrac{2x^{\frac{3}{2}}}{3}+C$

$=\dfrac{2\sqrt{3}\sqrt{x^3}}{3}+C=\dfrac{2\sqrt{3x^3}}{3}+C$

(i) $-\frac{28x^{\frac{5}{4}}}{5}+c$ index form

$-\frac{28\sqrt[4]{x^5}}{5}+c$ surd form

(j) $2x^{\frac{1}{2}}+c$ index form

$2\sqrt{x}+c$ surd form

(k) $-\frac{4x^{\frac{1}{2}}}{3}+c$ index form

$-\frac{4\sqrt{x}}{3}+c$ surd form

(l) $-\frac{2x^{-\frac{1}{2}}}{5}+c$ index form

$-\frac{2}{5\sqrt{x}}+c$ surd form

(m) $\frac{2x^{\frac{5}{2}}}{5}+c$ index form

$\frac{2\sqrt{x^5}}{5}+c$ surd form

(n) $\dfrac{dy}{dx}=4x^2\sqrt{x}=4x^2x^{\frac{1}{2}}=4x^{2+\frac{1}{2}}=4x^{\frac{5}{2}}$

$y=\dfrac{4x^{\frac{7}{2}}}{\frac{7}{2}}+c=4x^{\frac{7}{2}}\times\dfrac{2}{7}+c$

$=\dfrac{8}{7}x^{\frac{7}{2}}+c=\dfrac{8}{7}\sqrt{x^7}+c$

(o) $\frac{2x^{\frac{5}{2}}}{5}+c$ index form

$\frac{2\sqrt{x^5}}{5}+c$ surd form

Key Idea - Integration of Power of x where n is an Integer or Fraction

1. (a) $\frac{5x^2}{2}+\frac{x^3}{3}-\frac{3x^4}{4}+c$ (b) $\frac{x^3}{3}-x+c$ (c) $\frac{x^6}{6}-x^4-\frac{2x^3}{3}+x+c$

(d) $\frac{x^3}{15}-\frac{x^4}{16}+c$ (e) $\frac{2\sqrt{x^3}}{3}+\frac{3\sqrt[3]{x^4}}{4}+c$

(f)
$$\begin{aligned}\int -2x^4-\frac{1}{2x^2}\,dx &= \int -2x^4-\frac{1}{2}x^{-2}dx\\ &= \frac{-2x^{4+1}}{4+1}-\frac{1}{2}\times\frac{x^{-2+1}}{-2+1}+c\\ &= -\frac{2x^5}{5}-\frac{x^{-1}}{2\times(-1)}+c\\ &= -\frac{2x^5}{5}+\frac{1}{2x}+c\end{aligned}$$

(g) $\frac{-5}{x}+\frac{4\sqrt{x^3}}{3}+3x+c$ (h) $3x-\frac{1}{x}+\frac{1}{x^2}+c$ (i) $2\sqrt{x}-\frac{2}{\sqrt{x}}+c$

Key Idea - Integration by Simplifying the Integrand

1. (a)
$$\begin{aligned}\int 2x^2\left(3x+\frac{1}{x^2}\right)dx &= \int 6x^3+\frac{2x^2}{x^2}\,dx\\ &= \int 6x^3+2dx\,dx\\ &= \frac{6x^{3+1}}{3+1}+2x+c\\ &= \frac{3x^4}{2}+2x+c\end{aligned}$$

(b)
$$\begin{aligned}\int \left(x^2-5\right)\left(x^3-7\right)dx &= \int x^5-7x^2-5x^3+35\,dx\\ &= \frac{x^{5+1}}{5+1}-\frac{7x^{2+1}}{2+1}-\frac{5x^{3+1}}{3+1}+35x+c\\ &= \frac{x^6}{6}-\frac{7x^3}{3}-\frac{5x^4}{4}+35x+c\end{aligned}$$

(c)
$$\begin{aligned}\int \frac{2x^3-2x^5+x^9}{x^3}\,dx &= \int \frac{2x^3}{x^3}-\frac{2x^5}{x^3}+\frac{x^9}{x^3}\,dx\\ &= \int 2-2x^2+x^6\,dx\\ &= 2x-\frac{2x^{2+1}}{2+1}+\frac{x^{6+1}}{6+1}+c\\ &= 2x-\frac{2x^3}{3}+\frac{x^7}{7}+c\end{aligned}$$

(d)
$$\begin{aligned}\int 2x^3\left(3x^2-1\right)dx &= \int 6x^5-2x^3\,dx\\ &= \frac{6x^{5+1}}{5+1}-\frac{2x^{3+1}}{3+1}+c\\ &= \frac{6x^6}{6}-\frac{2x^4}{4}+c\\ &= x^6-\frac{x^4}{2}+c\end{aligned}$$

(e)

$$\int \frac{-2+x^4}{x^3}\,dx = \int \frac{-2}{x^3}+\frac{x^4}{x^3}\,dx$$

$$= \int -2x^{-3}+x\,dx$$

$$= \frac{-2x^{-3+1}}{-3+1}+\frac{x^{1+1}}{1+1}+c$$

$$= \frac{-2x^{-2}}{-2}+\frac{x^2}{2}+c$$

$$= x^{-2}+\frac{x^2}{2}+c$$

$$= \frac{1}{x^2}+\frac{x^2}{2}+C$$

(f)

$$\int \sqrt{x}\left(1+\frac{1}{\sqrt{x}}\right)dx = \int \sqrt{x}+1\,dx$$

$$= \int x^{\frac{1}{2}}+1\,dx$$

$$= \frac{x^{\frac{3}{2}}}{\frac{3}{2}}+c$$

$$= \frac{2x^{\frac{3}{2}}}{3}+c$$

$$= \frac{2\sqrt{x^3}}{3}+c$$

(g) answer only $\frac{4x^3}{3}+2x^2+x+c$

(h) answer only $\frac{x^5}{5}-x+c$